

MMA LAB QUIZ MASTER MCQ NOTE

CSE 306 - Microprocessor and Microcontroller Sessional

Strictly filtered to the uploaded lecture/lab syllabus

**226 UNIQUE MCQs | Slide-derived + web-inspired (paraphrased) |
Conceptual traps included**

Prepared for quiz practice - September 2026

How this master note was built

1. The uploaded slides and lab sheets define the syllabus boundary. Questions that require concepts not present in those files were excluded.
2. GeeksforGeeks, TutorialsPoint and Sanfoundry were used to cross-check standard 8086 terminology and to identify common MCQ patterns. Their questions were not copied verbatim; relevant ideas were rewritten and filtered against the class material.
3. Duplicate and near-duplicate prompts were removed. Many items are deliberately "trap-style": size mismatch, implicit registers, flag behavior, register-vs-memory, active-low display logic, and address/port mix-ups.
4. When a web page contains extra material outside the class slides (for example advanced 8255 modes), that material was not imported into the question bank.

Syllabus coverage map

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High-risk traps to memorize before the quiz

- 8086: 16-bit CPU, 20-bit address bus, 1 MB address space.
- Physical address = segment x 10H + offset.
- CL is the register used for variable shift/rotate counts in the taught syntax.
- TEST performs AND for flags but does not store the result; ZF=1 when that result is zero.
- LOOP automatically decrements CX and repeats while CX != 0.
- PUSH decrements SP by 2; POP increments SP by 2.
- Byte MUL: AX = AL x source. Word MUL: DX:AX = AX x source.
- Byte DIV: AX / src -> quotient AL, remainder AH. Word DIV: DX:AX / src -> quotient AX, remainder DX.
- Unsigned divide extension: clear AH/DX. Signed divide extension: CBW/CWD.
- Rotate preserves the bit pattern after a full-width rotation; logical shift discards bits.
- DOS INT 21H: AH=1 input -> AL, AH=2 character output from DL, AH=9 string output via DX and \$ terminator.
- 7-segment lab: common anode -> 0 lights a segment. Discrete LED lab: common cathode -> 1 lights an LED.
- Dot matrix: Port A green, Port B red, Port C columns; A+B gives orange.
- LCD BF=1 means busy; do not write the next instruction until BF=0.
- LCD line starts: 80H first line, C0H second line.

A. 8086 Architecture, Registers, Memory & Program Structure

1. The 8086 is described in the slides as a:

- A. 8-bit microprocessor
- B. 16-bit microprocessor
- C. 20-bit microprocessor
- D. 32-bit microprocessor

Answer: B | It can work on 16 bits at a time.

2. How wide is the 8086 address bus in the course slides?

- A. 8 bits
- B. 16 bits
- C. 20 bits
- D. 32 bits

Answer: C | A 20-bit address bus can address 2^{20} byte locations.

3. The maximum directly addressable memory implied by a 20-bit address bus is: **[TRAP]**

- A. 64 KB
- B. 256 KB
- C. 1 MB
- D. 16 MB

Answer: C | 2^{20} bytes = 1,048,576 bytes = 1 MB.

4. Which pair are the two major functional units of the 8086 CPU?

- A. ALU and RAM
- B. EU and BIU
- C. CS and DS
- D. AX and BX

Answer: B | The CPU is divided into the Execution Unit (EU) and Bus Interface Unit (BIU).

5. Which unit contains the ALU and performs arithmetic/logical operations?

- A. BIU
- B. EU
- C. ROM
- D. Stack segment

Answer: B | The EU contains the ALU and works with general registers.

6. Which unit communicates between the EU and memory/I/O?

- A. BIU
- B. ALU
- C. Flag unit
- D. Stack pointer

Answer: A | The BIU handles bus communication with RAM and I/O.

7. Which sequence best matches instruction execution in the slides?

- A. Decode -> Fetch -> Execute
- B. Fetch -> Execute -> Decode
- C. Fetch -> Decode -> Execute

D. Execute -> Fetch -> Decode

Answer: C | The slides list Fetch, Decode, Execute in that order.

8. How many total 16-bit registers are counted in the introductory slide?

- A. 8
- B. 12
- C. 14
- D. 16

Answer: C | The slide groups 4 data + 9 address + 1 flag register = 14.

9. Which register is most directly associated with loop counting?

- A. AX
- B. BX
- C. CX
- D. DX

Answer: C | CX is the count register and is used by LOOP.

10. Which register is emphasized for multiplication, division and I/O?

- A. AX
- B. BX
- C. SI
- D. BP

Answer: A | AX is the accumulator and is used for multiplication, division and I/O.

11. Which data register is also described as a base register?

- A. AX
- B. BX
- C. CX
- D. DX

Answer: B | BX is the Base Register.

12. Which data register is used with multiplication/division and output according to the register slide?

- A. AX only
- B. BX
- C. CX
- D. DX

Answer: D | DX is used in multiplication, division and output, especially for wider results/dividends.

13. Which list contains only segment registers?

- A. CS, DS, SS, ES
- B. SP, BP, IP, SI
- C. AX, BX, CX, DX
- D. SI, DI, CS, IP

Answer: A | The four segment registers are CS, DS, SS and ES.

14. Which list contains the pointer registers shown in the slides?

- A. SP, BP, IP

- B. SI, DI, IP
- C. CS, DS, ES
- D. AX, BX, CX

Answer: A | SP, BP and IP are listed as pointer registers.

15. Which are the index registers?

- A. AX and BX
- B. SI and DI
- C. SP and BP
- D. CS and DS

Answer: B | SI and DI are Source Index and Destination Index registers.

16. The 8086 physical address is formed by: [TRAP]

- A. segment + offset
- B. segment x 10H + offset
- C. offset x 10H + segment
- D. segment x 100H + offset

Answer: B | Shift the segment left 4 bits (multiply by 10H), then add offset.

17. What is the physical address of 1000H:0010H?

- A. 10010H
- B. 10001H
- C. 10100H
- D. 00110H

Answer: A | $1000H \times 10H = 10000H$; $+0010H = 10010H$.

18. Two different segment:offset pairs can refer to the same physical address because:

- A. segments are 8-bit
- B. segments can overlap
- C. offset is always zero
- D. BIU ignores segment values

Answer: B | Segmented addressing allows overlapping logical segments.

19. Which segment normally contains executable instructions?

- A. DS
- B. SS
- C. CS
- D. ES

Answer: C | CS is the Code Segment.

20. Which segment is used for stack data?

- A. CS
- B. DS
- C. SS
- D. ES

Answer: C | SS is the Stack Segment.

21. In the SMALL memory model used in the slides, the program has:

- A. many code and many data segments
- B. one code and one data segment
- C. one code and many data segments
- D. many code and one data segment

Answer: B | SMALL uses one code segment and one data segment, each up to 64 KB.

22. .STACK 100H reserves how many bytes? [TRAP]

- A. 100 bytes
- B. 128 bytes
- C. 256 bytes
- D. 512 bytes

Answer: C | 100H hexadecimal equals 256 decimal bytes.

23. What is the purpose of MAIN ENDP?

- A. Terminate DOS
- B. Mark the end of the MAIN procedure
- C. Initialize DS
- D. Declare stack size

Answer: B | PROC/ENDP delimit a procedure definition.

24. Why is END MAIN still needed after MAIN ENDP?

- A. It defines the data segment
- B. It tells the assembler the source end and entry point symbol
- C. It clears AX
- D. It creates a far procedure

Answer: B | END MAIN is an assembler directive marking the end and entry symbol; ENDP only closes the procedure.

25. Which memory model allows many code segments but only one data segment?

- A. SMALL
- B. MEDIUM
- C. COMPACT
- D. TINY

Answer: B | MEDIUM allows code >64 KB using many code segments while data stays <=64 KB.

B. Number Syntax, Variables, Basic Instructions & DOS I/O

26. Which is a valid binary literal according to the slides?

- A. 10110111
- B. 10110111b
- C. 0xB7
- D. B710

Answer: B | Binary literals end with b.

27. Which hexadecimal literal is valid?

- A. DAh
- B. 0DAh
- C. DA
- D. hDA

Answer: B | A hex literal beginning with A-F must be prefixed by a digit such as 0.

28. The literal 10110111 with no suffix is treated as:

- A. Binary
- B. Hexadecimal
- C. Decimal
- D. ASCII

Answer: C | Without b or h, the slide treats it as decimal.

29. Which variable name is illegal?

- A. @1
- B. AbC\$?
- C. ?
- D. 1Var

Answer: D | A variable name cannot start with a digit.

30. Assembly variable names in the slides are:

- A. Case-sensitive
- B. Case-insensitive
- C. Digits only
- D. Required to begin with @

Answer: B | The naming slide states case-insensitive.

31. DB defines a:

- A. 4-bit nibble
- B. 8-bit byte
- C. 16-bit word
- D. 32-bit double word

Answer: B | DB means Define Byte = 8 bits.

32. DW defines:

- A. 8 bits
- B. 16 bits
- C. 32 bits

D. 64 bits

Answer: B | DW means Define Word = 16 bits.

33. DD and DQ correspond to:

- A. 16 and 32 bits
- B. 32 and 64 bits
- C. 64 and 128 bits
- D. 8 and 16 bits

Answer: B | DD is 32 bits and DQ is 64 bits.

34. Why is VAR DB 320 invalid in the slide?

- A. 320 is negative
- B. 320 does not fit in an 8-bit byte
- C. DB requires binary only
- D. 320 is a register name

Answer: B | A byte can hold only an 8-bit value; 320 is too large.

35. Which instruction format is correct?

- A. Operand Opcode Comment
- B. Opcode Operands ;Comment
- C. Comment Operand Opcode
- D. Opcode ;Operand only

Answer: B | The slide shows opcode, operands, then semicolon comment.

36. What does MOV DEST, SRC do?

- A. DEST = DEST + SRC
- B. SRC = DEST
- C. DEST = SRC
- D. Swaps DEST and SRC

Answer: C | MOV copies the source value into destination.

37. Which MOV is illegal under the course restriction?

- A. MOV AH, DL
- B. MOV var1, AL when var1 is DB
- C. MOV var1, var2
- D. MOV AX, var1 when var1 is DW

Answer: C | The slides disallow both source and destination being memory variables at the same time.

38. Which MOV is illegal because of size mismatch?

- A. MOV AH, DL
- B. MOV AX, BX
- C. MOV CL, 1000000000000b
- D. MOV AL, 3

Answer: C | CL is 8-bit; that immediate does not fit the shown 8-bit use.

39. Which XCHG form is illegal?

- A. XCHG AH, DL
- B. XCHG var1, AL

- C. XCHG AX,30
- D. XCHG AX,BX

Answer: C | XCHG exchanges registers/memory, not an immediate constant.

40. ADD AL,BH means:

- A. $AL = BH$
- B. $AL = AL + BH$
- C. $BH = AL + BH$
- D. $AL = AL - BH$

Answer: B | ADD stores the sum in the destination.

41. SUB AL,BH means:

- A. $AL = AL - BH$
- B. $BH = BH - AL$
- C. $AL = BH - AL$
- D. Only flags change

Answer: A | SUB subtracts source from destination and stores the result in destination.

42. NEG AL performs:

- A. $AL = AL + 1$
- B. $AL = -AL$
- C. $AL = 0$
- D. $AL = \text{NOT } AL$ only

Answer: B | NEG forms the arithmetic negative (two's complement) of its operand.

43. INC AL is equivalent to:

- A. $AL = AL + 1$
- B. $AL = AL - 1$
- C. $AL = 1$
- D. $AL = AL + CF$

Answer: A | INC increments by one.

44. For DOS single-character input with INT 21H, AH should contain:

- A. 01H
- B. 02H
- C. 09H
- D. 4CH

Answer: A | AH=01H selects single-character input in the course slide.

45. After INT 21H with AH=1, the input character is returned in:

- A. AH
- B. AL
- C. DL
- D. DX

Answer: B | The input character is stored in AL.

46. For single-character output with INT 21H, the character must be placed in:

- A. AL

- B. AH
- C. DL
- D. DH

Answer: C | AH=2 selects output, and DL holds the character.

47. For DOS string output in the slide, AH should be:

- A. 1
- B. 2
- C. 9
- D. 10

Answer: C | AH=9 is the string-output service.

48. A string printed by the shown AH=9 service must end with:

- A. NUL
- B. CR
- C. \$
- D. #

Answer: C | The slide stores strings as DB data terminated by \$.

49. Before AH=9 string output, the string address is loaded into:

- A. AX using MOV
- B. DX using LEA
- C. CX using XCHG
- D. SP using PUSH

Answer: B | LEA DX,STRING loads the effective address into DX.

50. Why does the slide warn you to save a desired AL value before INT 21H? [TRAP]

- A. INT 21H always clears DS
- B. AL may be changed by the DOS service
- C. DX is always destroyed
- D. SP becomes zero

Answer: B | The slide explicitly warns that AL can change after INT 21H.

C. Control Flow, CMP, Conditional Jumps & Loops

51. Which instruction performs an unconditional transfer to a label?

- A. CMP
- B. JMP
- C. JE
- D. LOOP

Answer: B | JMP transfers control without testing a condition.

52. CMP is mainly used to:

- A. Store a subtraction result
- B. Update flags for a comparison
- C. Swap operands
- D. Push values

Answer: B | CMP compares values by subtraction-like flag updates without keeping the arithmetic result.

53. JE means:

- A. Jump if even
- B. Jump if equal
- C. Jump if error
- D. Jump if empty

Answer: B | JE branches when the comparison indicates equality.

54. JNE means:

- A. Jump if not equal
- B. Jump if negative
- C. Jump if no error
- D. Jump if no carry

Answer: A | JNE branches when values are not equal.

55. Which jump means greater?

- A. JG
- B. JL
- C. JGE
- D. JLE

Answer: A | JG is Jump if Greater.

56. Which jump means greater or equal?

- A. JG
- B. JGE
- C. JL
- D. JLE

Answer: B | JGE is Jump if Greater or Equal.

57. Which jump means less?

- A. JG
- B. JGE
- C. JL

D. JLE

Answer: C | JL is Jump if Less.

58. Which jump means less or equal?

- A. JG
- B. JGE
- C. JL
- D. JLE

Answer: D | JLE is Jump if Less or Equal.

59. JXCZ is used when:

- A. AX is zero
- B. BX is zero
- C. CX is zero
- D. CF is zero

Answer: C | JXCZ checks whether CX is zero.

60. JC and JNC test which flag?

- A. Zero
- B. Carry
- C. Sign
- D. Parity

Answer: B | JC jumps if carry; JNC jumps if no carry.

61. JO and JNO test:

- A. Overflow
- B. Zero
- C. Parity
- D. Sign

Answer: A | JO/JNO correspond to Overflow flag set/clear.

62. JP and JNP test:

- A. Pointer
- B. Parity
- C. Procedure
- D. Page

Answer: B | JP/JNP branch on the Parity flag.

63. JS and JNS test:

- A. Stack
- B. Sign
- C. Segment
- D. Shift

Answer: B | JS/JNS branch on the Sign flag.

64. The LOOP instruction uses which register automatically?

- A. AX
- B. BX

C. CX

D. DX

Answer: C | LOOP uses CX as its count register.

65. What does LOOP do to CX each iteration?

A. Increments it

B. Decrements it

C. Sets it to zero immediately

D. Rotates it

Answer: B | LOOP decrements CX and repeats while CX is not zero.

66. If CX starts at 5 and the loop body contains only LOOP TOP at the end, the body executes:

A. 4 times

B. 5 times

C. 6 times

D. Until CF becomes 0

Answer: B | The body executes once for each initial count value down to 1.

67. Why is the simple do-while translation guaranteed to execute its body at least once?

A. CMP is before the body

B. The condition is checked after the body

C. CX begins at zero

D. JMP always exits

Answer: B | A do-while checks its condition after executing the body.

68. To imitate a normal for-loop that might execute zero times, the condition should be checked:

A. Only after INC

B. Before entering/executing the body

C. Only after LOOP

D. Never

Answer: B | The slide's for-loop translation compares before the body so it can skip it.

69. In an assembly implementation of logical AND (&&), the first failed condition should usually:

[TRAP]

A. Still evaluate all code as true

B. Jump to the false/next branch

C. Set CF and continue true path

D. Call a procedure

Answer: B | Both conditions must be true, so any failure exits the true path.

70. In an assembly implementation of logical OR (|), a true first condition can usually:

A. Immediately enter the true branch

B. Force evaluation to false

C. Clear all flags

D. Skip the true branch

Answer: A | Only one OR term needs to be true.

D. Logical Operations, Bit Masks, Shift/Rotate & Binary/Hex I/O

71. AND DEST, SRC performs:

- A. Bitwise AND and stores in DEST
- B. Bitwise OR
- C. Only flag checking with no stored result
- D. Arithmetic addition

Answer: A | AND stores the bitwise AND result in destination.

72. OR is most useful for:

- A. Clearing selected bits
- B. Setting selected bits
- C. Dividing by 2
- D. Comparing signed numbers

Answer: B | OR with a 1 mask forces selected bits to 1.

73. AND with a 0 in a mask position will:

- A. Set that destination bit
- B. Clear that destination bit
- C. Toggle that bit
- D. Rotate that bit

Answer: B | Any bit AND 0 becomes 0.

74. XOR with 1 on a selected bit will:

- A. Keep it unchanged
- B. Clear it
- C. Toggle it
- D. Set CF only

Answer: C | XOR 1 complements that bit.

75. XOR with 0 on a bit will:

- A. Toggle it
- B. Leave it unchanged
- C. Always set it
- D. Always clear it

Answer: B | $X \oplus 0 = X$.

76. A common way to clear a register using only XOR is:

- A. XOR AL, 1
- B. XOR AL, AL
- C. XOR AL, 0FFH
- D. XOR AL, 32

Answer: B | Any value XOR itself becomes zero.

77. TEST differs from AND mainly because TEST:

- A. Does not perform a bitwise AND
- B. Does not store the AND result
- C. Never changes flags

D. Requires only one operand

Answer: B | TEST computes an AND for flag effects but does not store the result.

78. If TEST A,B produces an all-zero result, ZF becomes:

- A. 0
- B. 1
- C. Undefined
- D. Equal to CF

Answer: B | The Zero Flag is set when the result is zero.

79. To test whether AL is odd, the course uses:

- A. TEST AL,1
- B. TEST AL,2
- C. OR AL,1
- D. SHR AL,8

Answer: A | The least-significant bit determines odd/even.

80. After TEST AL,1, JZ EVEN is taken when:

- A. LSB=1
- B. LSB=0
- C. CF=1
- D. AL is negative

Answer: B | If the LSB is 0, the AND result is zero and ZF=1.

81. NOT performs:

- A. Two's complement
- B. One's complement
- C. Addition by 1
- D. Logical shift right

Answer: B | NOT flips all bits and does not add one.

82. NEG performs:

- A. One's complement only
- B. Two's complement arithmetic negation
- C. Rotate through carry
- D. Bitwise AND

Answer: B | NEG is two's complement: invert then add 1.

83. According to Week 9, which instruction does NOT affect flags?

- A. NEG
- B. NOT
- C. TEST
- D. AND

Answer: B | The slide says NOT does not affect any flags; NEG affects several.

84. SHL BL,1 sends the original:

- A. LSB to CF
- B. MSB to CF

- C. CF to MSB
- D. Zero Flag to LSB

Answer: B | One left shift moves the original MSB into CF.

85. SHR BL,1 sends the original:

- A. MSB to CF
- B. LSB to CF
- C. CF to LSB
- D. Parity flag to MSB

Answer: B | One right shift moves the original LSB into CF.

86. The vacant bit inserted by logical SHL is:

- A. 1 at MSB
- B. 0 at LSB
- C. CF at LSB
- D. Sign bit at LSB

Answer: B | Logical left shift inserts 0 at the right.

87. The vacant bit inserted by logical SHR is:

- A. 0 at MSB
- B. 1 at MSB
- C. CF at MSB
- D. Old LSB at MSB

Answer: A | Logical right shift inserts 0 at the left.

88. For the shift/rotate count operand in the course slides, the count must be: [TRAP]

- A. Any 8-bit register
- B. Only AL
- C. An immediate value or CL
- D. Only DX

Answer: C | The slide restricts the count to a value or CL.

89. Which is valid according to the shown shift-count rule?

- A. SHL BL,AL
- B. SHL BL,CL
- C. SHL BL,DL
- D. SHL BL,BH

Answer: B | CL is the register used for variable shift counts.

90. How many left shifts are sufficient to force any n-bit register to zero?

- A. n-1
- B. n
- C. 2n
- D. Always 1

Answer: B | After n logical left shifts, all original bits have been shifted out.

91. A one-bit left shift multiplies an unsigned value by 2 only when: [TRAP]

- A. Overflow/wraparound is ignored or does not occur

- B. CF is always 1
- C. The operand is a string
- D. CL is zero

Answer: A | Bit width can discard high bits; the multiplication equivalence assumes no lost significant bit.

92. ROL differs from SHL because ROL:

- A. Discards the MSB
- B. Feeds the shifted-out MSB back into the LSB
- C. Always inserts zero
- D. Does not use CF

Answer: B | ROL circulates the MSB to the LSB and also updates CF.

93. ROR BL,1 places the old LSB into:

- A. Only CF
- B. MSB and CF
- C. LSB and CF
- D. ZF and CF

Answer: B | ROR circulates the old LSB to MSB and copies it to CF.

94. After rotating an n-bit value left exactly n times, the value becomes:

- A. Zero
- B. All ones
- C. The original value
- D. The two's complement

Answer: C | A full-width rotation restores the original bit pattern.

95. Why is rotation convenient for binary output compared with destructive shifting?

- A. It can preserve/restore the original value after a full rotation
- B. It never changes CF
- C. It prints ASCII automatically
- D. It does not use loops

Answer: A | After n rotations the original pattern is restored, unlike shifts that discard bits.

96. RCL rotates through:

- A. Zero Flag
- B. Carry Flag
- C. Stack Pointer
- D. Instruction Pointer

Answer: B | RCL includes CF as an extra bit in the rotation path.

97. The course question about XORing a letter with decimal 32 is based on toggling:

- A. The ASCII case-related bit
- B. All 8 bits
- C. Only the sign flag
- D. The parity flag

Answer: A | Upper/lowercase ASCII letters differ by bit value 32 (20H).

98. Binary keyboard input is not directly a binary integer because the input arrives as:

- A. BCD words
- B. ASCII characters 0 and 1
- C. Hex nibbles only
- D. Raw carry flags

Answer: B | The slide explicitly treats binary input as a character string.

99. In the binary-input algorithm, before inserting each new bit, BX is:

- A. Shifted left
- B. Shifted right
- C. Rotated right
- D. Negated

Answer: A | Left shifting makes room at the LSB for the next bit.

100. The binary-input loop ends when the program reads:

- A. Space
- B. Dollar sign
- C. Carriage return
- D. Zero byte

Answer: C | The algorithm reads until CR.

101. For 16-bit binary output from BX, the slide repeats the output loop:

- A. 4 times
- B. 8 times
- C. 10 times
- D. 16 times

Answer: D | One bit is emitted per rotation, so 16 iterations print a 16-bit word.

102. In binary output using ROL BX, each next bit to print is obtained from:

- A. ZF
- B. CF
- C. PF
- D. AF

Answer: B | ROL moves the current MSB into CF.

103. For hexadecimal input, the register is shifted left by how many bits per input character?

- A. 1
- B. 2
- C. 4
- D. 8

Answer: C | Each hex digit represents 4 bits.

104. The simplified hex-input slide assumes at most:

- A. 2 hex characters
- B. 4 hex characters
- C. 8 hex characters
- D. 16 hex characters

Answer: B | It assumes no more than four uppercase hex characters.

105. For 16-bit hexadecimal output, the algorithm processes:

- A. 2 nibbles
- B. 4 nibbles
- C. 8 nibbles
- D. 16 nibbles

Answer: B | A 16-bit value contains four hexadecimal nibbles.

E. MUL/IMUL, DIV/IDIV & Decimal I/O

106. MUL performs:

- A. Signed multiplication
- B. Unsigned multiplication
- C. Unsigned division
- D. Signed division

Answer: B | MUL treats operands as unsigned.

107. IMUL performs:

- A. Signed multiplication
- B. Unsigned multiplication
- C. Signed division
- D. BCD conversion

Answer: A | IMUL is the signed multiply instruction.

108. For MUL BL, the implicit multiplicand is:

- A. AH
- B. AL
- C. AX
- D. DX

Answer: B | With an 8-bit source, $AX = AL \times \text{source}$.

109. For MUL BX, the implicit multiplicand is:

- A. AL
- B. AX
- C. DX
- D. CX

Answer: B | With a 16-bit source, $DX:AX = AX \times \text{source}$.

110. The product of two 8-bit operands is stored in:

- A. AL only
- B. AX
- C. DX only
- D. DX:AX

Answer: B | An 8x8-bit multiply can produce up to 16 bits, stored in AX.

111. The product of two 16-bit operands is stored in:

- A. AX only
- B. BX:AX
- C. DX:AX
- D. CX:DX

Answer: C | A 16x16-bit product can require 32 bits, stored in DX:AX.

112. After MUL, CF and OF are both 0 when:

- A. The upper half of the product is zero
- B. The lower half is zero
- C. The result is negative

D. ZF is one

Answer: A | For unsigned MUL, upper half zero means the lower half alone holds the result.

113. After IMUL, CF and OF are 0 when the upper half is:

- A. Always zero
- B. A correct sign extension of the lower half
- C. Equal to the divisor
- D. Equal to CF

Answer: B | That means the signed result fits in the lower destination half.

114. Which flags are emphasized as meaningful after MUL/IMUL?

- A. ZF and SF
- B. CF and OF
- C. PF and AF
- D. IF and DF

Answer: B | The slides focus on Carry and Overflow for multiplication.

115. DIV performs:

- A. Signed division
- B. Unsigned division
- C. Unsigned multiplication
- D. Bitwise division

Answer: B | DIV is unsigned division.

116. IDIV performs:

- A. Unsigned division
- B. Signed division
- C. Signed multiplication
- D. Logical right shift

Answer: B | IDIV performs signed division.

117. With an 8-bit divisor, the dividend is:

- A. AL
- B. AX
- C. DX
- D. DX:AX

Answer: B | Byte division divides AX by the 8-bit source.

118. With an 8-bit divisor, the quotient and remainder are stored in:

- A. AH quotient, AL remainder
- B. AL quotient, AH remainder
- C. AX quotient, DX remainder
- D. BL quotient, BH remainder

Answer: B | AL gets quotient and AH gets remainder.

119. With a 16-bit divisor, the dividend is:

- A. AX
- B. BX:AX

C. DX:AX

D. DS:AX

Answer: C | Word division uses the 32-bit dividend DX:AX.

120. With a 16-bit divisor, quotient and remainder go to:

A. AX quotient, DX remainder

B. DX quotient, AX remainder

C. AL quotient, AH remainder

D. BX quotient, CX remainder

Answer: A | AX receives quotient and DX receives remainder.

121. Before unsigned byte DIV when the original dividend is only in AL, the slide says to:

A. CBW

B. CWD

C. MOV AH,0

D. MOV DX,0

Answer: C | Zero-extend AL into AX by clearing AH.

122. Before signed byte IDIV when the dividend starts in AL, use:

A. MOV AH,0

B. CBW

C. CWD

D. XOR DX,DX

Answer: B | CBW sign-extends AL into AX.

123. Before unsigned word DIV when the original dividend is in AX, use:

A. MOV DX,0

B. CBW

C. CWD

D. MOV AH,0

Answer: A | Zero-extend AX into DX:AX by clearing DX.

124. Before signed word IDIV when the dividend is in AX, use:

A. MOV DX,0

B. CBW

C. CWD

D. NEG DX

Answer: C | CWD sign-extends AX into DX:AX.

125. CBW and CWD take:

A. One immediate operand

B. One register operand

C. No explicit operands

D. Two operands

Answer: C | They use fixed registers and specify no operands.

126. Flags after DIV/IDIV are:

A. All guaranteed zero

- B. Defined like CMP
- C. Undefined
- D. Only ZF defined

Answer: C | The Week 11 slide explicitly notes flags are undefined after division.

127. For decimal input, the running numeric total is updated as:

- A. $\text{total} = \text{total} + \text{digit}$
- B. $\text{total} = 2 * \text{total} + \text{digit}$
- C. $\text{total} = 10 * \text{total} + \text{digit}$
- D. $\text{total} = \text{digit} * \text{digit}$

Answer: C | Each decimal digit extends the previous value by one base-10 place.

128. For signed decimal input, the slide tracks a separate sign and applies which instruction at the end for negative input?

- A. NOT
- B. NEG
- C. ROL
- D. TEST

Answer: B | After building the magnitude, NEG is applied when sign is negative.

129. Decimal output extracts digits by repeatedly dividing by:

- A. 2
- B. 8
- C. 10
- D. 16

Answer: C | Division by 10 produces decimal digits as remainders.

130. Why are decimal-output remainders pushed onto the stack? [TRAP]

- A. To sort them
- B. They are generated least-significant first but must print most-significant first
- C. DIV requires stack operands
- D. To set CF

Answer: B | The stack reverses digit order for correct printing.

F. Stack & Procedures

131. The 8086 stack follows:

- A. FIFO
- B. LIFO
- C. Random access only
- D. Round-robin

Answer: B | The slides recap stack behavior as Last-In, First-Out.

132. PUSH SOURCE in the course slide accepts a:

- A. 8-bit byte only
- B. 16-bit register or memory word
- C. 20-bit physical address only
- D. String only

Answer: B | The slide specifies a 16-bit register or memory word.

133. What happens to SP on PUSH?

- A. SP increases by 1
- B. SP decreases by 1
- C. SP decreases by 2
- D. SP increases by 2

Answer: C | 8086 pushes a 16-bit word, so SP is decremented by 2.

134. What happens to SP on POP?

- A. SP decreases by 2
- B. SP increases by 2
- C. SP becomes zero
- D. SP changes by 1

Answer: B | Popping a word advances SP by 2.

135. The top of the stack is addressed using:

- A. CS:IP
- B. DS:SI
- C. SS:SP
- D. ES:DI

Answer: C | SS gives stack segment base and SP gives offset.

136. PUSH changes the source operand itself:

- A. Always
- B. Never; it copies the source
- C. Only for AX
- D. Only if CF=1

Answer: B | The source remains unchanged.

137. POP DEST moves data from:

- A. DS:SI
- B. SS:SP
- C. CS:IP

D. ES:BX

Answer: B | The content at the top of the stack is moved to destination.

138. Which destination is explicitly excluded for POP in the slide?

- A. AX
- B. CX
- C. DX
- D. IP

Answer: D | POP may target a 16-bit register except IP, or a memory word.

139. If AX=3, BX=2 and CX=1 are pushed in that order, the first value popped is:

- A. 3
- B. 2
- C. 1
- D. Undefined

Answer: C | LIFO returns the most recently pushed value first.

140. PUSHF and POPF operate on:

- A. Only AX
- B. Flags
- C. Only CS
- D. Only data variables

Answer: B | They push/pop the flags register state.

141. A procedure in the slides is best described as:

- A. A data segment
- B. A reusable named block of code
- C. A hardware interrupt
- D. A memory model

Answer: B | Procedures are reusable named code blocks called and returned from.

142. If PROC type is omitted, the slide says the procedure is assumed:

- A. FAR
- B. NEAR
- C. INTERRUPT
- D. EXTERN

Answer: B | NEAR is the default type.

143. A NEAR procedure is called from:

- A. A different segment
- B. The same segment
- C. Only ROM
- D. Only the stack segment

Answer: B | NEAR means caller and procedure are in the same segment.

144. A FAR procedure means the caller is:

- A. In a different segment
- B. In the same source line

- C. Always in DS
- D. Not allowed to return

Answer: A | FAR is used when the calling statement is in a different segment.

145. CALL name does which two key things in the slides?

- A. Clears stack and exits
- B. Stores return address on stack and changes IP to procedure
- C. Copies DS to ES and returns
- D. Only sets ZF

Answer: B | CALL saves where to return and transfers control to the procedure.

146. RET returns by:

- A. Popping the return address into IP
- B. Clearing AX
- C. Jumping to MAIN unconditionally
- D. Restoring DS from DX

Answer: A | RET gets the saved return address from the stack into IP.

147. Assembly procedures in the slides have built-in parameter lists like C functions:

- A. Yes, always
- B. No; parameters are arranged by the programmer
- C. Only FAR procedures do
- D. Only MAIN does

Answer: B | Values can be communicated through registers or the stack.

148. A procedure return value in the later array/procedure slide is commonly communicated through:

- A. A register such as AX
- B. The source code comment
- C. Only the flag register
- D. The .MODEL directive

Answer: A | The caller reads the result from an agreed register.

G. Arrays & Addressing Modes

149. The address assigned to the array variable itself is called the array's:

- A. Top address
- B. Base address
- C. Carry address
- D. Return address

Answer: B | The slide calls the address of the array variable the base address.

150. What does 100 DUP(0) mean in GAMMA DW 100 DUP(0)?

- A. One value 100
- B. 100 word elements initialized to zero
- C. 100 uninitialized bytes
- D. Duplicate address 100H

Answer: B | DUP repeats the given initializer the specified number of times.

151. DELTA DB 212 DUP(?) creates:

- A. 212 initialized words
- B. 212 uninitialized bytes
- C. 212 zeroed words
- D. A 212-bit register

Answer: B | DB makes bytes and ? means uninitialized storage.

152. ARRAY DB 5 DUP(?) is analogous to:

- A. Five initialized words
- B. An uninitialized 5-byte array
- C. One byte with value 5
- D. A 5-bit register

Answer: B | It allocates five byte elements without initialization.

153. LEA SI,ARRAY places into SI:

- A. arr[0] value
- B. The base/effective address of ARRAY
- C. The length of ARRAY
- D. The final element

Answer: B | LEA loads the address, not the element value.

154. MOV AL,[SI] means:

- A. Copy SI itself into AL
- B. Load the byte stored at the memory address pointed to by SI
- C. Store AL into SI register
- D. Increment SI

Answer: B | Brackets dereference the address in SI.

155. For a DB array, moving SI to the next element normally needs:

- A. INC SI
- B. ADD SI,2
- C. ADD SI,4

D. DEC SI twice

Answer: A | A byte element occupies one byte.

156. For a DW array, moving SI to the next element normally needs:

- A. INC SI
- B. ADD SI,2
- C. ADD SI,4
- D. SHL SI,4

Answer: B | A word occupies two bytes.

157. During LOOP-based traversal of 5 array elements, CX is typically initialized to:

- A. 0
- B. 1
- C. 5
- D. Array base address

Answer: C | CX holds the number of iterations/elements.

158. In register addressing mode, the operand is:

- A. A constant
- B. A CPU register
- C. A variable address written directly
- D. Always [BX]

Answer: B | Register mode names a register operand.

159. In immediate addressing mode, the operand is:

- A. A constant in the instruction
- B. A memory variable
- C. A pointer register only
- D. The stack top

Answer: A | Immediate mode includes the constant as part of the instruction.

160. In direct addressing mode as taught, the operand is specified by:

- A. A variable/memory location directly
- B. Only a register
- C. The stack pointer only
- D. A literal count only

Answer: A | Direct mode refers directly to a memory variable/address.

161. MOV AX,[SI] is an example of:

- A. Immediate
- B. Register indirect
- C. Direct register
- D. Based indexed with displacement

Answer: B | SI contains the offset address of the memory operand.

162. Which registers are allowed for register-indirect memory addressing in the slide?

- A. AX,CX,DX,SP
- B. BX,SI,DI,BP

- C. CS,DS,SS,ES
- D. AH,AL,BH,BL

Answer: B | The slide lists BX, SI, DI and BP.

163. When BP is used for indirect addressing, its default segment is: [TRAP]

- A. CS
- B. DS
- C. SS
- D. ES

Answer: C | The slide notes BP uses SS, while the others use DS.

164. When SI is used for indirect addressing, its default segment is normally:

- A. CS
- B. DS
- C. SS
- D. ES

Answer: B | SI normally uses DS according to the slide note.

165. An addressing mode using BX or BP plus displacement is called:

- A. Indexed
- B. Based
- C. Immediate
- D. Register only

Answer: B | Using a base register gives based addressing.

166. An addressing mode using SI or DI plus displacement is called:

- A. Based
- B. Indexed
- C. Immediate
- D. Stack

Answer: B | Using an index register gives indexed addressing.

167. [BX+SI] is an example of:

- A. Based indexed addressing
- B. Immediate addressing
- C. Direct addressing
- D. Only register addressing

Answer: A | It combines a base register and an index register.

168. Which instruction loads the value 100H into AX rather than the memory word at DS:0100H when SI=0100H? [TRAP]

- A. MOV AX,[SI]
- B. MOV AX,SI
- C. LEA AX,[SI]
- D. POP AX

Answer: B | Without brackets, MOV AX,SI copies the register value itself.

169. Which mode is especially useful for walking through a one-dimensional array using SI?

- A. Register indirect/indexed style
- B. Immediate only
- C. Flag mode
- D. Segment mode

Answer: A | The additional indirect/indexed modes are used for array processing.

170. Which statement about procedure parameters matches the array/procedure slide?

- A. They must be declared in parentheses
- B. They may be placed in registers or pushed on the stack
- C. Only AX can be used
- D. Procedures cannot receive values

Answer: B | The caller and procedure agree on registers/stack locations.

H1. MDA-8086 Trainer Kit

171. In the MDA-8086 trainer kit, KIT mode takes:

- A. Assembly source directly
- B. Machine language
- C. Only C code
- D. Python bytecode

Answer: B | The Lab 1 sheet says KIT mode takes machine language.

172. PC mode is described as taking:

- A. Only machine code
- B. Assembly or C programming language
- C. Only binary keypad input
- D. Only ROM data

Answer: B | The sheet states PC mode accepts assembly or C programming language.

173. Which keypad key performs a system reset?

- A. RES
- B. REG
- C. MON
- D. STP

Answer: A | RES is labeled system reset.

174. Which keypad key is used to set a memory address?

- A. AD
- B. DA
- C. GO
- D. MON

Answer: A | AD is labeled set memory address.

175. Which keypad key updates segment & offset and inputs data to memory?

- A. DA
- B. REG
- C. RES
- D. STP

Answer: A | DA is the data/memory update key described in the diagram.

176. Which key is described as register display?

- A. REG
- B. GO
- C. MON
- D. AD

Answer: A | REG is labeled Register Display.

177. Which key executes the user program one single step?

- A. GO
- B. STP
- C. RES

D. +

Answer: B | STP is labeled execute user's program, a single step.

178. Which key can go to the user program or execute monitor functions?

- A. MON
- B. GO
- C. REG
- D. DA

Answer: B | GO has that function in the keypad diagram.

179. Which key immediately breaks the user program and is associated with a non-maskable interrupt?

- A. MON
- B. STP
- C. AD
- D. RES

Answer: A | MON is the immediate break / non-maskable interrupt key.

180. The + and - keys are used mainly to:

- A. Change segment & offset/register display up or down
- B. Toggle LCD cursor
- C. Set CF
- D. Start DOS INT 21H

Answer: A | The keypad diagram labels increment/decrement of segment & offset/register display.

H2. LCD Interface with 8086

181. The LCD instruction-register I/O address in the lab manual is:

- A. 00H
- B. 02H
- C. 04H
- D. 1FH

Answer: A | 00H maps to the Instruction Register.

182. The LCD flag/status-register I/O address is:

- A. 00H
- B. 02H
- C. 04H
- D. 18H

Answer: B | 02H maps to the flag/status register.

183. The LCD data-register I/O address is:

- A. 00H
- B. 02H
- C. 04H
- D. 1AH

Answer: C | 04H maps to the Data Register.

184. The LCD controller has how many 8-bit registers emphasized in the manual?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B | It names the Instruction Register (IR) and Data Register (DR).

185. The Instruction Register (IR) is described as:

- A. Read-only
- B. Write-only
- C. 16-bit
- D. Used only for ASCII data

Answer: B | IR is write-only and stores instruction codes/addresses.

186. The Data Register (DR) stores:

- A. Only flags
- B. Data to be displayed, using character ASCII values
- C. Only cursor addresses
- D. Stack return addresses

Answer: B | DR holds display data; the manual says the data is ASCII value of the character.

187. For a 16x2 LCD, the first line address range is:

- A. 00H-0FH
- B. 40H-4FH
- C. 80H-8FH

D. C0H-CFH

Answer: C | The manual gives 80H through 8FH for line 1.

188. For a 16x2 LCD, the second line address range is:

- A. 10H-1FH
- B. 80H-8FH
- C. C0H-CFH
- D. F0H-FFH

Answer: C | The manual gives C0H through CFH for line 2.

189. In the line address, the second hex digit indicates:

- A. Row number only
- B. Column number 0 through F
- C. Busy state
- D. Interface width

Answer: B | The manual states the second digit indicates the column.

190. RS=0, R/W=0 selects:

- A. IR write / internal command operation
- B. Busy flag read
- C. DR write
- D. DR read

Answer: A | That combination writes the Instruction Register.

191. RS=0, R/W=1 selects:

- A. DR write
- B. Busy flag and address counter read
- C. DR read
- D. Display clear write

Answer: B | This reads BF on DB7 and address counter on DB0-DB6.

192. RS=1, R/W=0 selects:

- A. IR write
- B. Busy read
- C. DR write
- D. DR read

Answer: C | RS=1 selects data register; R/W=0 is write.

193. RS=1, R/W=1 selects:

- A. IR write
- B. Busy read
- C. DR write
- D. DR read

Answer: D | Both high means Data Register read.

194. Where does the LCD busy flag appear?

- A. DB0
- B. DB3

C. DB7

D. RS

Answer: C | BF is output on DB7 during RS=0 read.

195. If the busy flag is 1, the next instruction should:

A. Be written immediately

B. Wait until BF becomes 0

C. Toggle RS twice

D. Clear DR

Answer: B | BF=1 means an internal operation is in progress and new instructions are not accepted.

196. Which instruction takes about 1.64 ms in the table?

A. Clear display

B. Entry mode set

C. Display on/off control

D. Cursor/display shift

Answer: A | Clear display is shown with 1.64 ms execution time.

197. The clear-display command also returns the cursor to:

A. Column F

B. Home position/address 0

C. Second line

D. Last DDRAM address

Answer: B | The table says clear display returns the cursor home.

198. In the LCD bit table, I/D=1 means:

A. Decrement cursor

B. Increment cursor

C. Display off

D. Blink on

Answer: B | I/D selects cursor direction; 1 increments.

199. In the LCD bit table, S=0 means:

A. Shift display

B. No display shift

C. Cursor blink

D. 8-bit interface

Answer: B | S=0 is no display shift.

200. D=1, C=1, B=1 means the respective features are:

A. Display, cursor and blink all on

B. All off

C. Only display on

D. Only blink on

Answer: A | D, C and B are on/off controls for display, cursor and blink.

201. S/C=0 means:

A. Shift display

- B. Move cursor
- C. 8-bit mode
- D. Read DR

Answer: B | S/C=0 selects cursor movement; 1 selects display shift.

202. R/L=1 means:

- A. Shift left
- B. Shift right
- C. No shift
- D. Read status

Answer: B | R/L=1 selects right; 0 selects left.

203. DL=1 means:

- A. 4-bit interface
- B. 8-bit interface
- C. 1-line display
- D. 5x10 dots

Answer: B | DL selects interface width; 1 = 8-bit.

204. N=1 means:

- A. One line duty
- B. Two-line / 1/16 duty
- C. 5x10 font
- D. Busy

Answer: B | N=1 corresponds to 1/16 duty (2 lines).

205. F=1 means:

- A. 5x7 dots
- B. 5x10 dots
- C. Display off
- D. Cursor right

Answer: B | F=1 selects 5x10 dots.

206. BF=0 means:

- A. Internal operation in progress
- B. Can accept an instruction
- C. Display is off
- D. Data register selected

Answer: B | BF=0 indicates the LCD is ready for the next instruction.

H3. 7-Segment Display & Discrete LEDs

207. For the lab 7-segment setup, Port A address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1BH

Answer: B | Port A is 19H for the 7-segment display.

208. For the four discrete LEDs, Port B address is:

- A. 19H
- B. 1AH
- C. 1BH
- D. 1DH

Answer: C | Port B is 1BH.

209. The 7-segment setup Port C address is:

- A. 1BH
- B. 1CH
- C. 1DH
- D. 1EH

Answer: C | Port C is 1DH.

210. The control-register address for the 7-segment/LED setup is:

- A. 1CH
- B. 1DH
- C. 1EH
- D. 1FH

Answer: D | The control register is 1FH.

211. The 7-segment display in the lab is connected as:

- A. Common cathode
- B. Common anode
- C. Open collector only
- D. Matrix-only

Answer: B | The lab sheet explicitly says common anode.

212. For the common-anode 7-segment setup, a segment glows when its output bit is:

- A. 0
- B. 1
- C. Either always
- D. High impedance only

Answer: A | The sheet states 0 to glow segment LEDs.

213. Which mapping is correct for segment a?

- A. PA7
- B. PA3
- C. PA1

D. PA0

Answer: D | The table maps h g f e d c b a to PA7..PA0, so a=PA0.

214. Which mapping is correct for segment g?

A. PA6

B. PA5

C. PA1

D. PA0

Answer: A | g maps to PA6.

215. Which port drives the four standalone LEDs?

A. Port A

B. Port B

C. Port C

D. LCD DR

Answer: B | The sheet states Port B is used for LED.

216. The standalone LED connection is:

A. Common anode

B. Common cathode

C. No common terminal

D. Series only

Answer: B | The sheet labels the discrete LEDs as common cathode.

217. For the standalone common-cathode LEDs, a bit value of 1 will:

A. Turn the LED on

B. Turn the LED off

C. Reset the CPU

D. Select LCD IR

Answer: A | The sheet says 1 to glow LEDs.

218. Which LED is connected to PB2?

A. Left upper red

B. Green

C. Yellow

D. Bottom lower red

Answer: C | PB2 is the yellow LED.

H4. Dot-Matrix Interface

219. For the dot-matrix setup, Port A address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1CH

Answer: A | Port A is 18H.

220. For the dot-matrix setup, Port B address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1BH

Answer: C | Port B is 1AH.

221. For the dot-matrix setup, Port C address is:

- A. 1AH
- B. 1BH
- C. 1CH
- D. 1DH

Answer: C | Port C is 1CH.

222. The dot-matrix control-register address is:

- A. 1CH
- B. 1DH
- C. 1EH
- D. 1FH

Answer: C | Control register is 1EH.

223. In the dot-matrix sheet, Port A is connected to:

- A. Red LEDs
- B. Green LEDs
- C. Columns
- D. LCD data

Answer: B | Port A is connected with green LEDs.

224. In the dot-matrix sheet, Port B is connected to:

- A. Green LEDs
- B. Red LEDs
- C. Columns
- D. 7-segment common pin

Answer: B | Port B is connected with red LEDs.

225. In the dot-matrix sheet, Port C is connected to:

- A. Rows only
- B. Columns
- C. Only red LEDs

D. LCD RS

Answer: B | Port C is connected with columns.

226. Lighting both Port A and Port B LED colors together produces:

- A. Blue
- B. Orange
- C. White
- D. No light

Answer: B | The lab sheet states the combined color is orange.

Rapid Answer Key

Use this page for timed practice after covering the explanations.

1. B	24. B	47. C	70. A	93. B	116. B	139. C	162. B	185. B	208. C
2. C	25. B	48. C	71. A	94. C	117. B	140. B	163. C	186. B	209. C
3. C	26. B	49. B	72. B	95. A	118. B	141. B	164. B	187. C	210. D
4. B	27. B	50. B	73. B	96. B	119. C	142. B	165. B	188. C	211. B
5. B	28. C	51. B	74. C	97. A	120. A	143. B	166. B	189. B	212. A
6. A	29. D	52. B	75. B	98. B	121. C	144. A	167. A	190. A	213. D
7. C	30. B	53. B	76. B	99. A	122. B	145. B	168. B	191. B	214. A
8. C	31. B	54. A	77. B	100. C	123. A	146. A	169. A	192. C	215. B
9. C	32. B	55. A	78. B	101. D	124. C	147. B	170. B	193. D	216. B
10. A	33. B	56. B	79. A	102. B	125. C	148. A	171. B	194. C	217. A
11. B	34. B	57. C	80. B	103. C	126. C	149. B	172. B	195. B	218. C
12. D	35. B	58. D	81. B	104. B	127. C	150. B	173. A	196. A	219. A
13. A	36. C	59. C	82. B	105. B	128. B	151. B	174. A	197. B	220. C
14. A	37. C	60. B	83. B	106. B	129. C	152. B	175. A	198. B	221. C
15. B	38. C	61. A	84. B	107. A	130. B	153. B	176. A	199. B	222. C
16. B	39. C	62. B	85. B	108. B	131. B	154. B	177. B	200. A	223. B
17. A	40. B	63. B	86. B	109. B	132. B	155. A	178. B	201. B	224. B
18. B	41. A	64. C	87. A	110. B	133. C	156. B	179. A	202. B	225. B
19. C	42. B	65. B	88. C	111. C	134. B	157. C	180. A	203. B	226. B
20. C	43. A	66. B	89. B	112. A	135. C	158. B	181. A	204. B	
21. B	44. A	67. B	90. B	113. B	136. B	159. A	182. B	205. B	
22. C	45. B	68. B	91. A	114. B	137. B	160. A	183. C	206. B	
23. B	46. C	69. B	92. B	115. B	138. D	161. B	184. B	207. B	

Source & Provenance Notes

Primary syllabus authority: the 16 uploaded CSE 306 lecture/lab PDFs. The web was used only for MCQ pattern discovery and terminology cross-checking; the final bank was filtered back to what appears in the uploaded material.

Uploaded syllabus files

- Lecture 1(1).pdf
- Lec 2.pptx(1).pdf
- Lecture-03 (1).pdf
- Lec 4.pptx(1).pdf
- Lec 5.pptx(1).pdf
- Lec 6(1).pdf
- Lec 7(1).pdf
- Lec-8(1).pdf
- Lec-09.pdf
- CSE-306 Week 7,8.pdf
- CSE 306 Week-9.pptx.pdf
- CSE 306 Week 11.pptx.pdf
- Lab 1 (Intro) (1).pdf
- Lab 2 (LCD) (1).pdf
- Class 4 7-segment-LED (1).pdf
- DOT-Matrix Lab sheet (1).pdf

Web references used for MCQ-pattern research

- GeeksforGeeks - 8086 Instruction Set; Arithmetic Instructions in 8086; Flag Register of 8086; Programmable Peripheral Interface 8255; LCD Interfacing; Seven Segment Displays.
- TutorialsPoint - 8086 Overview; 8086 Functional Units; 8086 Addressing Modes; 8086 Instruction Sets; Memory Segmentation; Seven-Segment LED Display; related quiz pages.
- Sanfoundry - 8086 Addressing Modes MCQs; 8086/8088 Instruction Set MCQs; 8086/8088 Stack Structure MCQs; PIO 8255 MCQs.

Important note

"All possible MCQs" cannot literally be finite because a teacher can generate unlimited new numeric values and code variants. This note instead provides a broad, non-duplicate coverage bank of every concept visible in the supplied syllabus, with extra emphasis on common mistake patterns.